Changes in the New 5th Edition of API 616 Specifications and Their Implication for Gas Turbine Pipeline Compression Applications

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ABSTRACT

API 616 forms the backbone for most gas turbine driver purchases and station design integration in the oil and gas industry. This year, the 5th edition was introduced to replace the 4th edition which had been in place since 1998. This paper provides a summary overview of the 5th edition of American Petroleum Institute (API) Code API 616 and also provides some comparison to the 4th edition. Some critical differences between the 4th and 5th edition are highlighted and discussed. Relevant new sections of the code are discussed in more detail with a special focus on their technical interpretation and relevance for the scope-of-supply comparison, machine testing, and field operation in pipeline service. Some recommendations for acceptance of manufacturer exceptions to API and the technical/ commercial implications are also provided.

INTRODUCTION

API specifications are generally applied to oil and gas turbomachinery applications rather than large industrial power generation. Oil and gas applications of gas turbines have requirements that are inherently different than those of the electric power industry. Namely, oil and gas applications and customers require:

- High Availability/ Reliability
- Ruggedness
- High Power/ Weight Ratio
- Low Cost of O&M

Because of these inherent market differences, oil and gas customers often insist on compliance with API codes and are willing to accept the resulting higher turbomachinery package costs.

Many applications within the oil and gas industry require the usage of turbomachinery equipment for compression, pumping, and generation of electricity. These applications are generally divided into three separate areas from the production of oil to the sales of the refined product. These are upstream, midstream, and downstream. Upstream refers to the oil and gas

production and gathering process, midstream is the oil and gas transportation process, and downstream is the refining and distribution of oil and gas products. Although API 616 is applicable to all gas turbines, this paper primarily focuses on midstream applications; i.e., pipeline natural gas compression service. Codes and standards are used by turbo compressor manufacturers and users to specify, select, and characterize their equipment. Codes are generally used for the following purposes:

- Provide design guidelines and minimum standards
- Facilitate and simplify procurement
- Provide high level quality assurance
- Assure system interconnectivity and compatibility.

However, no code "designs" a gas turbine, and codes are by nature rather general. Any particular application may require modifications. The applicability of API 616 to oil & gas application and individual components of gas turbines was discussed by Brun (2006), and is beyond the scope of this paper.

For turbomachinery applications in the oil and gas sector, the most commonly used standards are:

- API 616 – Gas Turbines
- API 617 – Centrifugal Compressors
- API 614 – Lube Oil System
- API 670 – Machinery Protection
- API 613 – Load and Accessory Gear
- API 677 – Accessory Drive Gear
- API 671 – Flexible Couplings
- NFPA 70 – Electric Code
- ASME PTC-22 – Gas Turbine Testing
- ASME PTC-10 – Compressor Testing

Of the above codes, API 616 and 617 are the most critical codes when evaluating or purchasing a turbo compressor package. However, API 614 and 670 are strongly referenced in both API 616 and 617, and thus, should also be critically reviewed when purchasing a gas turbine driven turbo compressor. A critical review of API 616, 4th edition, was provided by Brun (2006).

WHAT IS API?

The American Petroleum Institute (API) is the primary trade organization for the U.S. petroleum industry. API has over 400 member companies that cover all aspects of the oil and gas production. API is an accredited American National Standards Organization (ANSI) and started developing industry specific codes in 1924. Currently, API publishes about 500 standards, which

are widely referenced by EPA, OSHA, BLM, ASME, and other codes and regulations. API's philosophy for developing codes is based on the following principals:

- Improve Safety
- Improve Environmental Performance
- Reduce Engineering Costs
- Improve Equipment Interchangeability
- Improve Product Quality
- Lower Equipment Cost
- Allow for Exceptions within Reason

However, as with most other codes, API specifications often lack technology developments, especially in the rapidly changing gas turbine and compressor markets. The 4th edition of API 616 was released in August 1998, and the 5th edition was released in January 2011. It took almost 12 years to update this standard.

API STANDARDS AS PURCHASING SPECIFICATIONS

API standards are often used as a convenient purchasing document. They provide the means for a customer to normalize the quotations by forcing all machinery vendors to quote on similar scopes. API codes also provide a common language and set of rules between vendor and customer to limit misunderstandings (e.g., definitions of efficiencies, test procedures, vendor data, data sheets).

All API codes clearly state in their foreword that exceptions are allowed, if they lead to an improved or safer technical offer. For gas turbine applications in the oil and gas sector, API 616 is the foundation for almost all purchase specifications. API 617 is used for most centrifugal compressor applications. Also, API references the National Fire Protection Agency (NFPA) NFPA 70 electric code for hazardous locations. This specification is fundamental in most machinery standards and is, thus, a critical requirement for oil and gas applications. Namely, oil and gas turbocompressors must at least meet Class 1, Division 2 (Zone 2), Group D. Some applications require Division 1 (Zone 1).

API STANDARD TOPICS

The API 616 code can be divided by a set of standard topics. These are:

Foreword (Philosophy)

1. **Scope (and alternative designs)**
2. **References (more or less everything is referenced)**
3. **Definitions**
    - ISO rating, normal operating point, maximum continuous speed, trip speed, etc.

- Note: Some basic requirements are hidden here (e.g. MCS)

4. **Basic Design**
   - Pressure ratings, rotordynamics, bearings/ seals, balancing requirements, materials, lubrication
   - Covers quality and mechanical integrity issues
   - Primarily core engine
5. **Accessories**
   - Starters, inlet/ exhaust, mounting, fuel, gears, enclosures, fire protection, tools
   - Mostly on-skid package items
6. **Inspection, Testing, and Preparation for Shipment**
   - Required and optional tests: hydrostatic, mechanical run, package, PTC 22
   - Long term and short term shipping
   - Minimal test requirements
7. **Vendor Data**
   - Drawings, performance data, calculations, quality documentation
   - References, Appendix B list.
8. **Appendix**
   A. Data sheets
   B. Vendor drawing and data requirements
   C. Procedure to determine residual unbalance (balancing)
   D. Lateral and torsional logic diagrams
   E. Gas turbine nomenclature

Gas turbine ancillaries and auxiliaries and their relevance in API 616 were discussed in detailed in Brun (2010).

Often overlooked, but critically important, are the API data sheets, as they clearly form the technical basis of a typical proposal. Within these data sheets, the application specific issues are addressed, such as customer site, operating conditions, basic equipment selection, and equipment minimum integrity requirements. Thus, for a purchaser of a turbo compressor, it is important to always fill out (as a minimum) the data sheets for API 616 (Gas Turbine), API 617 (Compressor), API 614 Appendix D (Lube Oil System), and API 670 Appendix A (Machinery Protection). When filling out these data sheets, a couple of industry accepted norms should be remembered:

- Cross out requirements that do not apply or cannot be complied with.
- Include notes for critical technical comments. By including them on the data sheets, these comments are elevated to a contractual technical requirement.

- Fill out as much information as is available: Even a partially filled out sheet is better than no sheet.

API 616 FIFTH EDITION

After a 5-year effort involving approximately 60 gas turbine expert engineers, the API released in January 2011 the fifth edition of API 616 (Gas Turbines for Petroleum, Chemical, and Gas Industry Services). Since the fourth edition of the API 616 standard was originally released in August 1998, this was long overdue and clearly necessary to catch up with gas turbine technology. The new fifth edition corrects many of the problems of the fourth edition and is very much an improvement. There are hundreds of changes in the fifth edition. It is beyond the scope of this paper to address all these changes, but this paper discusses some of the more controversial items. Specifically, there are new paragraphs and changes in the fifth edition that are at least worth some further analysis and evaluation. Since one of the authors of this paper (Brun) was a member of the fifth edition API taskforce, and thus, is, at least partially responsible for its content, the authors will refrain from overly negative editorializing.

The authors emphasize that a careful review of API specification applicability should be performed for every gas turbine purchase based on the specifics of the application. Also, the reader should note that API 616 is often considered as not applicable to Aero-Derivative Gas Turbines (i.e., it is intended for industrial type gas turbines only), but that in reality, there is nothing within API 616 that excludes this type of turbines. As a matter of fact, API 616 5th edition deliberately accommodates Aero-Derivative Gas Turbine.

Thus, a limited number of the more interesting and relevant topics of the new API 616 5th edition are discussed below:

4.3.7 Manufacturer must indicate Wobbe index rate of change capability

In the previous edition of API 616, the OEM only had to indicate the gas turbine's operable Wobbe index range. Additionally, there was, and still is in the 5th edition, a requirement that the gas turbine can handle a heating value change of +/-10% (5.8.2.4). However, the new edition not only requires that the manufacturer states the allowable Wobbe index range, but he must also state the allowable rate of change within this range. This is particularly critical in pipeline service where a "slug" of natural gas can travel down a pipeline from a well or a source of LNG with widely varying fuel properties. Most gas turbines can handle a fairly wide range of Wobbe index (often above +/-10%), but their control system's response time is limited and may have problems dealing with rapid changes in LHV. By including an API requirement that manufacturers have to indicate the allowable rate of change of Wobbe index, the purchaser can now estimate whether a gas turbine will have operational issues or even flame-outs in a particular installation or application.

4.5.1.2 Gas Turbine Shafts can be stacked, welded drum, or solid

The fourth edition of API had, for some unexplainable reason, a requirement that all gas turbine rotors must be of solid construction. This is interesting, since there are very few gas turbine manufacturers who build their rotors this way. Almost all gas turbine rotors are stacked disks,

and there are some frame units that use a welded drum construction. Most likely, this requirement came out of an earlier edition of API 617, which had this specification and was intended solely for centrifugal compressors. Fortunately, the new edition of API 616 has this problem corrected and now allows gas turbine shafts to be stacked, welded-drum, or solid rotors.

4.7.4 High speed balancing is allowed

Previous editions of API 616 did not allow for high speed balancing and required that a rotor and its subcomponents (disks) should only be low speed balanced. The new edition of API 616 reverses this and allows high speed balancing. This is somewhat troubling, as this opens the door for machines with flexible rotors that cannot be brought to an acceptable unbalance with low speed balancing and require to be high speed balanced. At first glance, this may appear reasonable since most manufacturers have high speed balancing capabilities. The problem with this is that it eliminates the possibility of field-repairing a machine, since users generally do not have high speed balancing machines (which are very complex and expensive). There are actually only a limited number of high speed balancing machines available in the world for large rotors. I.e., accepting machines that require high speed balancing will often result in the requirement that a rotor is sent to the original equipment manufacturer's factory for all rotor repairs (or individual blade replacements).

4.7.5 Specifies vibration limits of gas turbines similar to API 617 and ISO codes

Similar to API 617 (Centrifugal Compressors) and ISO 7919 (Rotating Shafts), API 616 now has a requirement for vibration limits for gas turbines. The vibration limits are set differently for industrial and aeroderivative gas turbines, since on industrial gas turbines, absolute vibrations are usually measured with proximity probes, whereas on aeroderivative units, case vibrations are measured with seismic probes. Although having vibration limits may appear reasonable, it really does not make a lot of sense for all gas turbines, especially units that have significant fleet experience. Specifically, if a gas turbine model has demonstrated hundreds of thousands of vibration trouble free hours over many units in the field, it makes little sense to impose artificial vibration requirements. By requiring the OEMs to meet these limits, they may actually be forced to change the dynamics of the system (e.g., bearing stiffness or damping) which could, in some cases, negatively impact the life of the gas turbine. I.e., if it ain't broke, don't fix it.

5.3.2.1 Multiple bolted baseplates are permitted

In the fourth edition of API 616, the driver and driven equipment had to be mounted on a single piece solid baseplate. This was very difficult to meet in cases when the gas turbine and compressor were supplied by different companies as each had their own baseplate design specifications. Also, for some very large gas turbine driven compressor trains, a single base plate is not feasible for size and weight transport limitations. Thus, most manufacturers in the past took exception to this, and the two baseplates were tightly bolted at their interfaces. This fundamentally works and has not resulted in significant operational issues. In a few cases, primarily in offshore application, a sub-base can be mounted under the bolted baseplates of the equipment to avoid transmission of platform surface twisting and bending moments. The fifth edition of API 616 recognized that this approach works well and does allow for separate bolted

baseplates, as long as the purchaser agrees. If multiple baseplates are used, their interfaces must be machined and doweled to allow for accurate field assembly.

5.4.2.2.2 Control system verifies unit purge before startup

Past API specification required a certain amount of time of purging the gas turbine exhaust system prior to start up. This safety requirement is necessary to avoid having any flammable concentrations of fuel and air in the exhaust system that could ignite during the startup. The requirement was based on replacing multiple times the total stored volume of air in the entire exhaust system prior to engaging the gas turbine start-up sequence. However, in a few well documented cases, the control system was based on a timed purge cycle with the assumption that the gas turbine crank would provide the required air flow. However, in some cases, the crank did not engage or the crank motor failed to start such that the air in the exhaust was never replaced prior to startup. To eliminate this risk, the new edition of API requires that the control system verifies the gas turbine compressors shaft speed during the crank cycle to assure proper purge flow.

5.6 A new, very, very detailed discussion on design and inlet system

API standards typically refrain from "designing" equipment and usually provide minimum technical requirements only. However, the new API 616 section on inlet filters provides a detailed design guideline for gas turbine inlet filtration including such specifics as number of filtration stages, type of filter in each stage, weather protection, instrumentation, etc. This is unfortunate, as past experience has shown that inlet filtration is very site and application specific, and that there is no single filtration design that is appropriate for all conditions. For example, although API now requires a three-stage filter with weather protection for all applications, this is certainly not necessary for gas turbine installations in clean industrial environments. Similarly, the high or low velocity filter combinations, as specified in detail, will not be adequate for some offshore or desert applications. It would have been better, had API specified the required filtration efficiency and protection against water ingress for a given pressure drop rather than designing the inlet filtration system.

5.6.1.10 Inlet system must be stainless steel

API 616 5th edition requires that the entire inlet system, including all dusting, housing, transition pieces, etc., are made from stainless steel. This can easily add several hundreds of thousands of dollars of cost to every new gas turbine unit. Clearly, stainless steel is better than carbon steel, and for salty air environments, is strongly recommended. However, in many applications in a dry environment, such as in a desert location, painted carbon steel is fully adequate. A couple of buckets of paint cost far less than stainless steel. There is really no reason to make this a firm API requirement, as it may unnecessarily add cost to a new installation. The necessity of this should have been left to the purchaser to determine.

5.6.1.16 Power turbine cleaning system

The new edition of API 616 requires that the gas turbine is supplied with a compressor and power turbine cleaning system. This appears to be a mistake since gas turbine suppliers will usually only provide compressor on-crank and on-line washing, but very few can actually

provide a power turbine washing system. Off-line power turbine washing is only required in some specialized applications using fuels that are extremely dirty such as heavy crude oils and refinery “bottom-of-the-barrel” residuals. Very few gas turbines on the market are equipped with power turbine cleaning capabilities, and this should not be an API requirement.

5.8.1.2.2 Manual fuel shutoff valve outside package required

The fifth edition of API 616 has added a requirement for a manual fuel shot-off valve that is strategically located away from the gas turbine package, so that in case of a fire, the gas supply can be easily shut-off. This is a sensible requirement that is in line with almost all fire codes, and most gas turbine facilities already have it installed. It is important to note that most gas turbine manufacturers will point out that they are not responsible for off skid fuel gas piping and that this manual valve is really the responsibility of the purchaser or the provider of the balance of the compression plant.

5.8.2.5.1 Customer must provide gas composition to C12 for fuel

Liquid drop out of heavy hydrocarbons in a gas turbine’s fuel gas system from the natural gas is a major cause for gas turbine combustor and hot section turbine failures. Previously, the gas analysis that was provided to the manufacturer to determine the dew point suitability of a fuel and to determine the need for fuel gas heating was based on limited gas analysis, typically up to C8 or less. The new edition of API 616 corrects this problem by requiring that the purchaser provide a gas analysis up to C12. Based on this, the manufacturer should be able to better determine the actual requirement for fuel gas heating to achieve the necessary superheat of the fuel to avoid liquid drop outs in the fuel system. One note of caution: In many applications the original design fuel gas is not the same as the fuel the gas turbine actually sees when installed in the facility. I.e., although a gas composition up to C12 is available, this may not solve the problem of liquid drop out during actual operation of the gas turbine.

5.8.1.2.7 Feedback monitor for fuel valve in close position

The critical fuel valve control and shut-off valve on the gas turbine skid are assumed to be properly functioning. For the fuel control valve, this is usually verified through the measurement of the fuel gas pressure downstream of the valve and through the indirect measurement of the firing temperature. However, the fuel control valve and shut-off valve may not fully close when a unit is down, which can potentially lead to a dangerous leak of natural gas into the gas turbine, inlet system, and exhaust system. To avoid this, the fifth edition of API 616 has a new requirement that there is a feedback monitor of the fuel valves to the control system. A valve that is not fully closed should lead to an immediate alarm and blocking of any startup attempts.

6.2.2.1 Material examination “equivalent international tests” are allowed

The new API 616 5th edition is somewhat vague on the material examination testing requirements with a statement that “Equivalent International Test Standards” are acceptable. Past API standards required that materials meet ASTM standards, and most US operators and manufacturers are very familiar with these. The subject wording was clearly used to allow for European and Japanese manufacturers to meet API requirements, but API should have been more specific on which international standards are acceptable. As it stands, any standard from

any country, even if it is completely inadequate, would be acceptable. Furthermore, the quality control in some international test laboratories is often questionable. This allowance could potentially lead to poor materials being utilized in the fabrication of key gas turbine components and could lead to catastrophic failure events.

6.3.3 Optional additional pneumatic pressure test

API 616 in the past included a mandatory hydro pressure test that meets ASME requirements for all pressure vessels. The new edition of API also includes an optional pneumatic pressure test of critical components. This test is useful for testing complete systems that are usually only tested as individual components. For example, the fuel gas system and the dry gas seal system are typically not pressure tested after assembly (only individual pipes and pieces are tested) and operators often have to deal with small gas leakage issues during commissioning activities. By performing a pneumatic pressure test of these systems in the factory, this costly delay in the field could often be avoided.

6.3.5.2 Defines "Full Load String Test"

The new note in this section of the fifth edition provides really nothing new but it defines for the first time the term "Full Load String Test." This term has been used through the industry for many years, but nobody really knew what the specific test requirements were. API 616 5th edition now states that the Full Load String Test is simply a combination of a complete unit test and a performance test. The complete unit test is the 4-hour mechanical run test that has always been required by API 616, but includes driven equipment, gears, starter motor, auxiliary equipment, etc. The performance test is fundamentally based on ASME PTC 22 or ISO 2314. It does not include a closed loop compressor test (ASME PTC 10). Thus, if a purchaser requires a full load string test, it should now be clear to the manufacturer how to perform the test and what equipment is included. This should avoid a lot of confusion on this topic in the future.

6.3.5.5 Purchaser can request 120% rotor overspeed test

The 120% overspeed testing of a rotor would appear reasonable, as this would provide some safety margin in case of a sudden loss of load on the gas generator shaft. However, it is difficult to determine if this 20% overspeed requirement is based on some real physical behavior of a gas turbine, or if it simply is an arbitrary safety margin. During a sudden load loss event, such as a coupling burst, some small gas turbines may exceed 20%, while large frame units will probably stay well below this value. The amount of acceleration a turbine experiences during such an event requires a complex calculation that depends on the inertia of the rotor, the remaining fuel downstream of the fuel control valve, the time it takes to sense an overspeed event, and the ability of the control system to react to this overspeed event. Thus, simply adding a 20% margin does not really provide an added safety margin, if it is unknown how the turbine will react. This requirement also may lead to overdesigned rotors as it forces manufacturers to meet the 120% overspeed specification, even if it is not reasonable.

6.3.5.7/8 Optional enclosure ventilation and leak test

The fifth edition of API 616 includes a new optional test to determine enclosure leakage and proper ventilation. A ventilation test usually does not. The leak test is of limited value as all

enclosures are designed to allow for a certain leakage rate, and the fire fighting CO_2 or water mist release system is designed to overcome this leakage rate. I.e., an enclosure usually leaks, and a leak test will usually only confirm this.

6.3.5.9 Documented borescope inspection required after mechanical run test

Previous, API 616 4th edition required that the gas turbine unit is disassembled after the mechanical run test and then reassembled. This obviously makes no sense whatsoever. Why would you disassemble and reassemble a perfectly functional unit and increase the risk of making a mistake in the process that results in a failure? The fifth edition of API corrected this problem and only requires a borescope inspection of the hot gas path after a successful test. The borescope photos must then provided to the purchasers so that they have documentation of the baseline (from the factory) condition of the unit for comparisons with later borescope inspections in the field. However, if the unit fails the mechanical run test, API 616 still requires disassembly of the unit, but specifically states that this is done to determine the reason for the unit not passing the test. This is an overall more sensible approach.

6.3.6 Optional field test per PTC 22 or ISO2314 or GMRC Guidelines

API 616 5th edition now provides for the selection by the purchaser an optional field test of the gas turbine unit. In many cases, this is done to determine whether a unit meets contractual performance guarantees and, thus, has been offered by all manufacturers as a standard option. By including this as an option in the new API 616 and defining which test standards are acceptable, it makes it easier for purchasers to include this early in the project planning (and costing) process.

CONCLUSIONS

API 616 forms the backbone for most pipeline gas turbine driver purchases and station design integration. This paper provides a summary overview of the recently released fifth edition of API 616 and also provides some comparison to the fourth edition.

Although API 616 5th edition is a clear improvement over the fourth edition, there are still a number of items that do not appear to make sense for pipeline applications. This paper described some of the more interesting and, possibly, more controversial changes of the new edition. Based on the above comments, the following recommendations are provided to pipeline operators regarding the applicability of API 5th edition requirements to their gas turbine compression equipment:

Section 4.3.7 requires that the manufacturer provide the allowable fuel Wobbe index rate of change to the purchaser for the gas turbine. This is very valuable information, and the pipeline operator should take advantage of this data. In pipeline applications, the Wobbe index can rapidly change due to transient supply conditions. Thus, it is imperative that the purchaser review and analyze the allowable Wobbe rate of change data to verify whether the gas turbine may encounter operational problems for the desired pipeline compression application, accounting for all current and future operating scenarios, prior to its installation in the plant.

Section 4.7.4 allows the usage of gas turbines that require high speed balancing for oil & gas applications. This is not very good advice for pipeline compressor operators, though. Because of their fairly flexible rotor designs, these machines are difficult to repair on site and often cannot even be properly trim-balanced. They must usually be shipped to a high speed balancing facility after any rotor repair or wheel/ blade replacement. This is clearly inconvenient and inefficient for pipeline compression where high equipment availability is critically important.

Section 5.6.1.10 states that the complete gas turbine inlet system must be made from stainless steel. For most onshore pipeline compression applications, this is a very costly overdesign. A properly applied multi-coating paint job is usually sufficient in non-salty air environments and requires minimal maintenance. Regular touch up painting and inspections should be performed as necessary, but this is still usually less expensive than a complete stainless steel inlet system.

Section 5.8.1.2.7 discusses reporting of gas composition up to C12 requirements. This is important, and complying with this requirement can avoid serious gas turbine operational, safety, and part life problems in the future. The pipeline operator should always provide the gas composition for all foreseeable future operating conditions to the manufacturer up to C12 (or higher, if possible) during the early phases of any new station design project.

The rotor overspeed test required in Section 5.3.5.5 is probably something pipeline operators should not insist on, especially for any gas turbine models that have a large number of units successfully operating in the field. This should only be considered for prototype or first-of-a-kind units.

Section 6.3.5.7/8 is an optional leak and ventilation test for the enclosure. Since most pipeline units employ standard enclosures and package features, it is unlikely that this test will provide any added safety for a pipeline station operator. These types of test make sense for one-of-a-kind gas turbine packages or maybe for novel firefighting technology testing, but not for standardized gas turbine packages that are typically encountered in pipeline compression service and that have hundreds of identical fleet units in similar applications.

Section 6.3.6 provides an option for a field performance test of the gas turbine. Given that high efficiency and energy utilization have become a critical mandate for pipeline operators, this test should be performed to verify whether the gas turbine meets guarantee performance on site. The test results also provide a gas turbine and compressor performance baseline to determine and trend future degradation over time.

The above are just generic recommendations that may not be applicable for all types of operations or site requirements. As always, a careful review of the specification should be performed by anyone purchasing a gas turbine and, if necessary, exceptions from the manufacturer should be allowed based on the site condition and application.

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